A Digital-Twin Multi-Agent Framework for Enterprise Cognitive Networking: ECNetBench and Leakage-Safe Multi-Seed Evaluation

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***Abstract*—Enterprise cognitive networking—combining digital twins, predictive analytics, explainable root-cause analysis (RCA), and closed-loop remediation support—lacks a shared, leakage-safe benchmark that couples realistic multi-layer enterprise configuration with multi-modal telemetry and supervised task labels. We present *ECNetBench*, a synthetic enterprise benchmark grounded in AOS-CX-style operational assumptions, together with an *Enterprise Cognitive Network* (ECN-v3) comprising Perception, Anomaly, Prediction, RCA, Impact, and Healing agents. The final T1 detector uses leakage-safe temporal feature enrichment with *anchored* telemetry-first fusion (`ECNFusionModel`); nesting-safe stacking is evaluated as an ablation and does not improve mean T1 AUPRC. RCA uses random-forest classification with TreeSHAP explanations. Evaluation follows a temporal $70/15/15$ protocol on six frozen instances (v1.1.0-INST and seeds 101–505). Across seeds, the final ECN-v3 detector attains mean T1 AUPRC of 0.1152 (bootstrap 95% CI $[0.0679, 0.1680]$), exceeding random forest (0.0758) and the prior v2 configuration (0.0577). The recommended T2 head is telemetry logistic regression (0.0380). Exact floating-point values are archived in `results/manuscript_ready_numbers.json`. We do not claim universal superiority on every paired test (n=6). Instead, we contribute (i) a reproducible multi-seed benchmark with leakage rules, (ii) a complete detection–prediction–RCA–impact–healing decision-support pipeline with an evidence-selected fusion head, and (iii) honest multi-seed confidence intervals, calibration, and paired significance tests.**

***Index Terms*—Enterprise networking, digital twin, multi-agent systems, anomaly detection, failure prediction, root-cause analysis, AIOps, benchmark, reproducibility.**

I. Introduction

Modern enterprise campuses and data-center fabrics are operated under continuous change: VLAN and ACL updates, EVPN/VXLAN overlays, QoS policy edits, and firmware-driven control-plane behavior. Cognitive NetOps and AIOps promise earlier anomaly detection, failure-horizon prediction, explainable RCA, service-impact estimation, and remediation recommendations [1]–[3]. Beyond purely technical detectors, enterprise modernization also depends on governance maturity, workforce adaptation, and cybersecurity readiness when operators adopt digital twins and automation [4]–[7]. In practice, progress is slowed by fragmented public resources: topology archives without multi-modal telemetry [8], security flow corpora that omit enterprise L2/L3 configuration semantics [9], and telemetry/RCA artifacts that do not jointly expose recovery and impact labels under a temporal leakage protocol.

This paper addresses that gap with two coupled contributions. First, **ECNetBench** is a synthetic enterprise cognitive-networking benchmark whose schema, failure taxonomy, and label definitions are designed for leakage-safe supervised learning under AOS-CX-style operational assumptions (inventory, VLAN/ACL/QoS, routing, counters, syslog/alerts, NAE-like analytics, and recovery/impact fields). Second, we implement and evaluate an **Enterprise Cognitive Network (ECN-v3)**: a Digital Twin graph plus a multi-agent stack (Perception, Anomaly, Prediction, RCA, Impact, Healing) with leakage-safe feature enrichment, *anchored* telemetry-first fusion (`ECNFusionModel`), and TreeSHAP-explained RCA.

a) Novelty statement.: We contribute an integrated twin + multi-agent detection–prediction–RCA–impact–healing *decision-support* pipeline evaluated on a frozen multi-seed enterprise benchmark under an explicit temporal $70/15/15$ leakage protocol with paired Wilcoxon tests and Cliff's δ across six independent instances. We do not claim absolute historical priority over all related AIOps or digital-twin systems; the novelty is the coupled benchmark, leakage protocol, and evidence-selected fusion evaluation.

b) Contributions.:

1) **ECNetBench design:** relational+graph schema, failure taxonomy, causal incident modeling, leakage-safe labels, realism validation, and six frozen evaluable instances (v1.1.0-INST; seeds 101, 202, 303, 404, 505).
2) **ECN-v3 framework:** Digital Twin construction; leakage-safe temporal enrichment; anchored fusion selected over nesting-safe stacking by architecture isolation; RF+TreeSHAP RCA.
3) **Rigorous multi-seed evaluation:** baselines, fusion ablations, calibration, confidence intervals, significance tests, and computational cost—with honest reporting of non-significant paired tests where n=6 is underpowered.
4) **Reproducible package:** relative-path evaluation harness, checksums, traceability artifacts, and a synchronized Overleaf project under `paper/overleaf/`.

c) Headline results (exact verified means, n=6).: Final ECN-v3 T1 AUPRC = 0.1152 (bootstrap 95% CI $[0.0679, 0.1680]$) versus strongest baseline random forest = 0.0758 and versus the prior v2 configuration = 0.0577. Nesting-safe stacking on the same enriched features is a *negative* ablation (0.0997). The recommended T2 head is telemetry logistic regression (0.0380). Twin contribution under the final T1 head is near zero ($\approx +0.00035$).

The overall architecture is introduced in Fig. 2 (Section IV). Section II situates related work; Sections III–IV detail the benchmark and framework; Sections V–VI present protocol and results; Sections VII–IX discuss implications, limitations, and conclusions.

II. Related Work

A. Network digital twins and autonomous networks

Network digital twins (NDTs) aim to provide data-driven, near-real-time models for what-if analysis and closed-loop control [3], [10]. Standards bodies and industry white papers discuss NDTs as enablers of autonomous networks and safe change validation [11], [12]. Our twin focuses on enterprise campus/DC inventory and telemetry graphs rather than packet-level WAN simulation, and is evaluated as a feature provider inside a multi-agent NetOps pipeline.

B. AIOps, anomaly detection, and RCA

Surveys of AIOps and network anomaly detection highlight class imbalance, concept drift, and the need for operationally meaningful metrics beyond accuracy [1], [2], [13]. Classical detectors (Isolation Forest [14], EWMA) and strong tabular learners remain competitive when features are informative. RCA research spans log mining, causal graphs, and Clos-fabric diagnostics [2]. We evaluate explainable category-level RCA with ablations that remove twin or syslog cues, and we treat healing as *decision support* rather than live switch actuation.

C. Graph learning for networks

Graph neural networks and message-passing models have been proposed for topology-aware prediction [15], [16]. We evaluate both a historical GraphSAGE-style *proxy* (LightGBM on a restricted feature subset; retained for continuity) and a **true** GraphSAGE mean-aggregator over the Digital Twin device graph. Attention-tabular models such as TabNet [17] are included as a modern deep tabular baseline. The proposed ECN uses an explicit Digital Twin with neighbor aggregates and anchored fusion rather than claiming end-to-end GNN or TabNet superiority.

D. Benchmarks and datasets

Public resources include Topology Zoo [8], SNDlib [18], and security datasets such as UNSW-NB15 [9]. Enterprise cognitive NetOps additionally needs configuration semantics, multi-modal telemetry, service impact, and recovery labels under temporal freeze rules. ECNetBench is synthetic by design; we therefore emphasize leakage audits, seed robustness, and explicit threats to external validity (Section VIII).

E. Intent-based and multi-agent network management

Intent-based networking and multi-agent orchestration are long-standing themes in automated management [19], [20]. Recent surveys of agentic and multi-agent systems emphasize tool use, evaluation benchmarks, and governance frameworks for delegated autonomy [21], [22]. ECN-v3 instantiates specialized agents with an orchestrator; fusion is constrained (anchored telemetry weight ≥ 0.5) after architecture isolation showed nesting-safe stacking does not improve mean T1 AUPRC on ECNetBench. RCA explanations use TreeSHAP on a random-forest category model.

TABLE I
Qualitative comparison of ECNetBench against common public resource classes for enterprise cognitive NetOps research.

Resource class	Config.	Telemetry	RCA/heal	Multi-seed
Topology archives	Partial	No	No	Rare
Security flow sets	Low	Flows	Limited	Varies
Clos/telemetry RCA	Fabric-spec.	Yes	RCA-focused	Varies
ECNetBench	**Yes**	**Yes**	**Yes**	**Yes**

F. Governance, cybersecurity, and operational trust

As NetOps pipelines become more automated, enterprise deployments inherit broader digital-transformation and cybersecurity concerns: risk controls in IT organizations [23], trust-oriented auditing of operational records [24], and zero-exposure practices for identity-recovery workflows [25]. Parallel literature on enterprise generative and agentic AI highlights security/governance risks and prioritization architectures for retrieval-augmented or generative assistants [26], [27]. These works motivate treating ECN healing outputs as governed decision support rather than unsupervised actuation, and motivate leakage-safe evaluation and artifact traceability as scientific analogues of operational auditability.

III. ECNetBench: Benchmark Design

A. Design goals

ECNetBench targets enterprise cognitive networking research with five goals: (G1) AOS-CX-style multi-layer configuration realism; (G2) multi-modal telemetry (counters, resources, syslog/alerts, flow aggregates); (G3) causal incident chains linking failure, detection, recovery, and impact; (G4) leakage-safe supervised labels; (G5) multi-seed robustness under frozen instances. Table I contrasts ECNetBench with common public resource classes for cognitive NetOps research.

B. Enterprise assumptions and topology

Instances model a compact enterprise fabric (19 devices, 31 links in the evaluated family) spanning campus/DC roles with VLAN membership, ACL/QoS bindings, routing (e.g., BGP/OSPF-related tables), LAG/VSX-like redundancy concepts, and overlay-related objects where applicable. Topology snapshots and typed graph edges support Digital Twin construction. Generation constraints and schema documents accompany the release.

C. Telemetry and causal incident modeling

Synthetic generators emit interface counters, device resources, environmental/power samples, syslog/alert streams, and service KPIs. Failure incidents are drawn from a taxonomy covering link, device, control-plane, policy, and service-impacting families. Each incident carries detection, recovery-action, and impact fields so that labels for anomaly windows, failure horizons, RCA categories, impact, and healing actions remain consistent with a causal narrative. Fig. 1 summarizes the generation and validation workflow from constrained synthesis to audited SQLite instances and supervised labels.

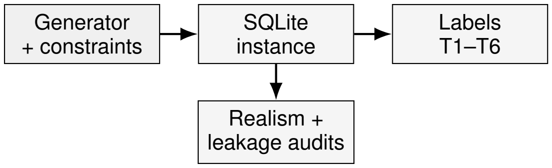

Fig. 1. ECNetBench generation and validation workflow: constrained synthesis produces a frozen SQLite instance that is audited for realism/leakage and exported to supervised task labels.

TABLE II
FROZEN EVALUATION INSTANCES USED IN THIS STUDY (AUTHORITATIVE SQLITE STORES).

Instance folder	Role	Devices / links
`v1` (v1.1.0-INST)	Primary frozen instance	19 / 31
`v1.1-seed101`	Independent seed	19 / 31
`v1.1-seed202`	Independent seed	19 / 31
`v1.1-seed303`	Independent seed	19 / 31
`v1.1-seed404`	Independent seed	19 / 31
`v1.1-seed505`	Independent seed	19 / 31

TABLE III
SUMMARY OF ECNETBENCH RELATIONAL SCHEMA FAMILIES.

Family	Representative contents
Inventory/topology	Devices, interfaces, topology snapshots, graph nodes/edges
Configuration	VLAN/ACL/QoS, routing processes, configuration snapshots/diffs
Telemetry	Interface counters, resources, syslog/alerts, flow aggregates
Services	Applications, services, SLA and impact bindings
Incidents/labels	Failure incidents, recovery actions, supervised label tables

D. Leakage-safe labels and splits

Labels are aligned to time bins. Features at decision time t_0 use only information available at or before t_0 (historical shifts; no future counters). Evaluation uses a temporal 70/15/15 train/validation/test split frozen per instance. Forbidden fields (raw future labels, oracle incident IDs in feature matrices) are audited in the companion leakage reports.

E. Instances and seed robustness

We evaluate six frozen SQLite instances: `v1.1.0-INST` and independent seeds 101, 202, 303, 404, and 505, as listed in Table II. SHA-256 digests are published in `benchmark/INSTANCE_CHECKSUMS.json`. Seed-robustness and publication-validation reports accompany the package. SQLite is authoritative for evaluation; CSV/Parquet exports are optional derivatives. Schema families are summarized in Table III.

IV. ENTERPRISE COGNITIVE NETWORK FRAMEWORK

A. Digital Twin

The Digital Twin materializes inventory and topology as a typed graph synchronized with telemetry windows. Twin-derived features include residual contrasts (e.g., device resource versus neighbor summaries), structural aggregates, and interface error/discard/carrier deltas constructed with causal shifts. Twin features are optional inputs to agents and are ablated in `no_twin` / `twin_only` settings.

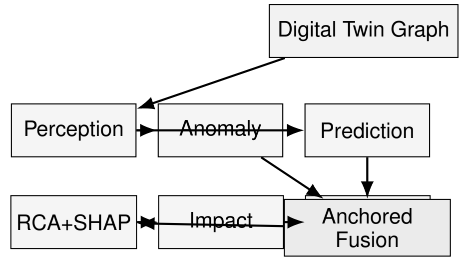

Fig. 2. ECN-v3 multi-agent workflow: Perception feeds Anomaly/Prediction specialists into *anchored* orchestrated fusion, followed by TreeSHAP RCA, impact estimation, and healing decision support.

B. Leakage-safe feature enrichment (ECN-v3)

Beyond legacy aggregates, ECN-v3 adds causal rolling/EMA statistics, second differences, error-burst accumulators, and neighbor-instability proxies. All rolling and expanding operators apply `shift(1)` so that the current bin is excluded; labels join features at $\texttt{feat\_bin} = t_{\text{start}} - 30\,\text{min}$. Isolation studies attribute the T1 gain primarily to this enrichment rather than to fusion complexity.

C. Multi-agent architecture

Fig. 2 depicts the agent workflow. Perception builds leakage-safe feature matrices from telemetry and twin views. Anomaly and Prediction score T1 anomaly windows and T2 failure-horizon risk, respectively. RCA predicts failure category with TreeSHAP explanations (T3); Impact estimates service-impact labels (T4); and Healing recommends remediation classes as *decision support* (TR-AUTO). Actions are not executed on physical switches in this study.

a) Final fusion head.: Architecture selection under the frozen six-seed protocol compares nesting-safe stacking against anchored fusion on the same enriched features. Anchored fusion (`ECNFusionModel`) constrains the telemetry specialist weight to be at least 0.5 (or selects telem-only), yielding higher mean T1 AUPRC with a simpler operator-facing interpretation. Stacking (`ECNStackFusionModel`) is retained as an ablation/negative result. The recommended T2 head is telemetry logistic regression under ultra-rare positives.

D. Class imbalance and calibration

Positive rates are low ($\sim$1% for T1; lower for T2). Training uses imbalance-aware weighting where applicable (`scale_pos_weight`); Isolation Forest scores are mapped via a train empirical CDF. Thresholds for F1/precision/recall are tuned on validation only and are *not* used for AUPRC. Primary ranking metrics are AUPRC and ROC-AUC; Brier score reports calibration. Optional Beta/Platt calibration may be applied for probability display without changing AUPRC ranking.

V. Evaluation Protocol

A. Tasks

We report T1 anomaly detection, T2 failure-horizon prediction, T3 RCA (macro-F1 with TreeSHAP), T4 impact, and TR-AUTO healing decision support. Supplementary tasks (degradation, config risk) are produced by the harness and archived in `results/`.

B. Baselines and fairness

Baselines include logistic regression, random forest, LightGBM, XGBoost, CatBoost, gradient boosting, balanced random forest, Isolation Forest, EWMA, threshold rules, majority class, an MLP sequence proxy, **TabNet** [17] (telem_only), a historical GraphSAGE-style *proxy* implemented as LightGBM on a restricted feature subset, and a **true GraphSAGE** [15] message-passing model over the Digital Twin device adjacency (pure PyTorch mean aggregator; node features = telemetry + light twin scalars per device–time bin). Classical tabular baselines train on telemetry-only feature matrices; the proposed T1 model uses twin+telemetry with **anchored** fusion on leakage-safe enriched features. Shared temporal enrichment is applied consistently so comparisons are not confounded by unequal feature engineering for baselines versus proposed specialists. We explicitly distinguish the historical GNN *proxy* from the true GraphSAGE baseline; neither replaces the final ECN-v3 T1 head.

C. Metrics and statistics

Primary metric: average precision (AUPRC). Secondary: ROC-AUC, precision, recall, F1 (validation-tuned threshold), Brier score. Aggregation: mean and 95% CI across six seeds. Significance: paired Wilcoxon signed-rank tests and Cliff's δ of the final T1 head versus each baseline and versus the stacking ablation on per-seed AUPRC. With many paired comparisons and n=6, p-values are interpreted as exploratory evidence rather than family-wise controlled discoveries; we do not apply multiplicity correction and therefore do not claim simultaneous significance across the baseline suite.

D. Ablations

Primary architecture ablations contrast (i) v2 legacy features + anchored fusion, (ii) v3 enriched features + stacking, and (iii) v3 enriched features + anchored fusion (final). Module deltas report feature-enrichment gain versus v2, twin contribution under the final head, and stacking versus anchored on identical features.

E. Computational cost

Full-suite wall-clock time with TabNet and true GraphSAGE averages approximately 120 s per seed (six-seed re-run including deep baselines; see `results/scalability_measured.json`). Mean T1 training time for the anchored head remains $\approx$ 2.2 s; TabNet and GraphSAGE training are substantially slower ($\approx$ 16 s each on CPU).

VI. Results

All numerical claims below are taken from verified artifacts `results/manuscript_ready_numbers.json`, `results/v3_gated/t1_architecture_selection.json`, and `results/aggregate_v3.json` (baselines/T2/RCA), without modifying frozen instances.

A. T1 anomaly and T2 failure prediction

Table IV summarizes multi-seed mean ranking metrics. On T1, the final ECN-v3 anchored detector achieves mean AUPRC 0.1152 (bootstrap 95% CI $[0.0679, 0.1680]$; architecture-selection artifact `results/manuscript_ready_numbers.json`), exceeding random forest (0.0758) and the v2 legacy configuration (0.0577). Mean T1 ROC-AUC is 0.7534. Fig. 3 visualizes the same AUPRC means with 95% confidence intervals for T1 and T2, highlighting the T1 lift of the final head and the T2 telem-logistic recommendation.

On T2, the recommended head is telemetry logistic regression with mean AUPRC 0.0380, higher than RF (0.0176). Table IV's ECN T2 columns report the multi-agent proposed head ($\approx$ 0.0153), which is *not* the recommended operating head. We do not claim that multi-agent fusion improves T2 under ultra-rare positives.

Representative ROC and precision–recall curves on the primary frozen instance appear in Fig. 4. Curve geometry remains consistent with rare-positive ranking: ROC discrimination is moderate, while precision–recall is challenging for all methods. Legends label the harness **anchored** ECN-v3 head (`model_family=anchored_v3`); instance-level AP/ROC values need not equal the six-seed selection mean.

Paired Wilcoxon tests and Cliff's δ (Table V) show that the T1 mean advantage versus random forest is not significant at α=0.05 (p=0.688, δ=+0.389), consistent with n=6 underpowering. Versus logistic and several weaker baselines, paired p-values reach 0.031 (exploratory under multiple comparisons; Section V). Stacking versus anchored is not significant (p=0.375).

B. Architecture selection and fusion ablation

Fig. 5 and Table VI isolate the sources of the T1 improvement. Leakage-safe feature enrichment accounts for the dominant gain; replacing anchored fusion with stacking on the same enriched features *reduces* mean AUPRC from 0.1152 to 0.0997. Accordingly, stacking is reported as an ablation/negative result rather than as the final T1 architecture.

Table VII and Fig. 6 summarize contribution deltas under two complementary definitions. Feature enrichment versus v2 contributes +0.0575 mean AUPRC (architecture selection). Twin gain under the final head is near zero (+0.00035). Stacking relative to anchored contributes −0.0155. Separately, the harness module table reports a slightly negative twin ablation delta (−0.0043) under the latest full-suite re-run; both estimates support the same qualitative conclusion that Twin is not the primary T1 ranking driver.

TABLE IV

Multi-seed mean detection metrics on T1 (anomaly) and T2 (failure horizon). Entries are means over six frozen instances; AUPRC is reported with 95% confidence intervals. The ECN T1 AUPRC/ROC use the authoritative final selection in `manuscript_ready_numbers.json`. ECN T2 columns report the multi-agent proposed head (not the recommended telem-logistic operating head; see Logistic row). Other rows (including TabNet and true GraphSAGE) use the latest `aggregate_v3.json` re-run.

Method	T1 AUPRC	T1 ROC-AUC	T2 AUPRC	T2 ROC-AUC
ECN (proposed)	**0.1152** [0.0679, 0.1680]	**0.7534**	**0.0153** [0.0043, 0.0262]	**0.6371**
TabNet	0.0492 [0.0213, 0.0772]	0.6300	0.0110 [0.0038, 0.0183]	0.6149
GraphSAGE (true)	0.0152 [0.0059, 0.0246]	0.5653	0.0062 [0.0057, 0.0068]	0.5340
XGBoost	0.0471 [0.0234, 0.0707]	0.7010	0.0115 [0.0069, 0.0161]	0.6039
CatBoost	0.0528 [0.0174, 0.0882]	0.6703	0.0117 [0.0049, 0.0184]	0.5710
Gradient boosting	0.0730 [0.0500, 0.0961]	0.7590	0.0129 [0.0082, 0.0175]	0.7077
Balanced RF	0.0467 [0.0192, 0.0743]	0.7751	0.0176 [0.0049, 0.0302]	0.7392
Random forest	0.0758 [0.0427, 0.1090]	0.7661	0.0176 [0.0099, 0.0253]	0.7081
Logistic	0.0631 [0.0235, 0.1028]	0.7260	0.0380 [−0.0027, 0.0788]	0.6891
LightGBM	0.0570 [0.0262, 0.0879]	0.6798	0.0213 [−0.0041, 0.0467]	0.5483
Isolation Forest	0.0575 [0.0255, 0.0895]	0.6716	0.0216 [−0.0040, 0.0472]	0.6459
EWMA	0.0403 [0.0135, 0.0671]	0.5484	0.0066 [0.0045, 0.0086]	0.4482
GNN proxy	0.0335 [0.0167, 0.0504]	0.6846	0.0106 [0.0069, 0.0144]	0.5965
MLP sequence	0.0084 [0.0053, 0.0115]	0.3617	0.0054 [0.0041, 0.0067]	0.3795
Majority	0.0098 [0.0083, 0.0113]	0.5000	0.0056 [0.0050, 0.0061]	0.5000

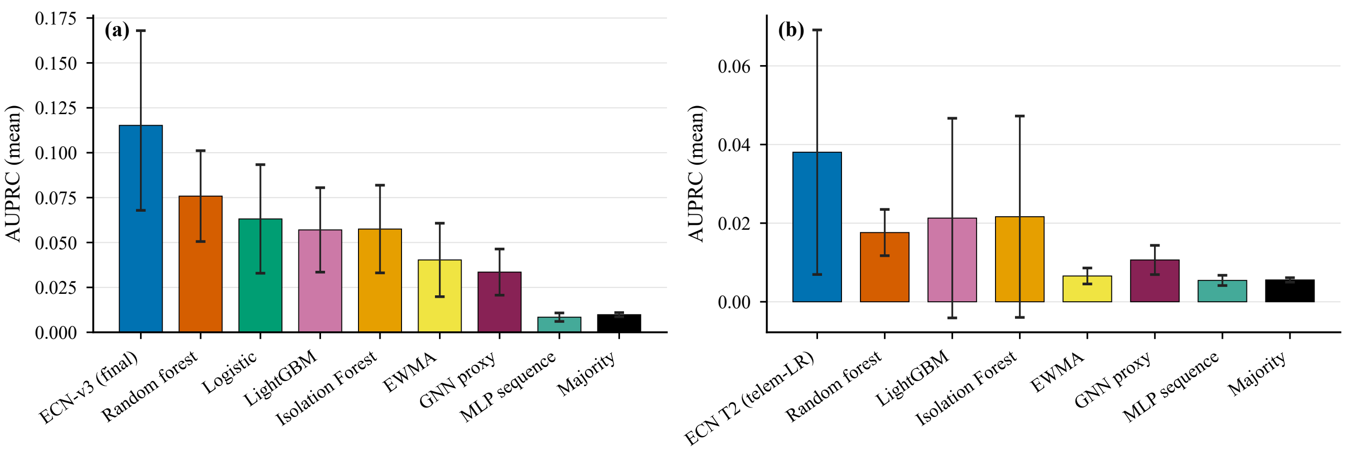

Fig. 3. Multi-seed mean AUPRC with 95% confidence intervals. (a) T1 anomaly detection, where ECN-v3 denotes the final anchored head on enriched features. (b) T2 failure-horizon ranking; the ECN T2 column is the recommended telemetry logistic head.

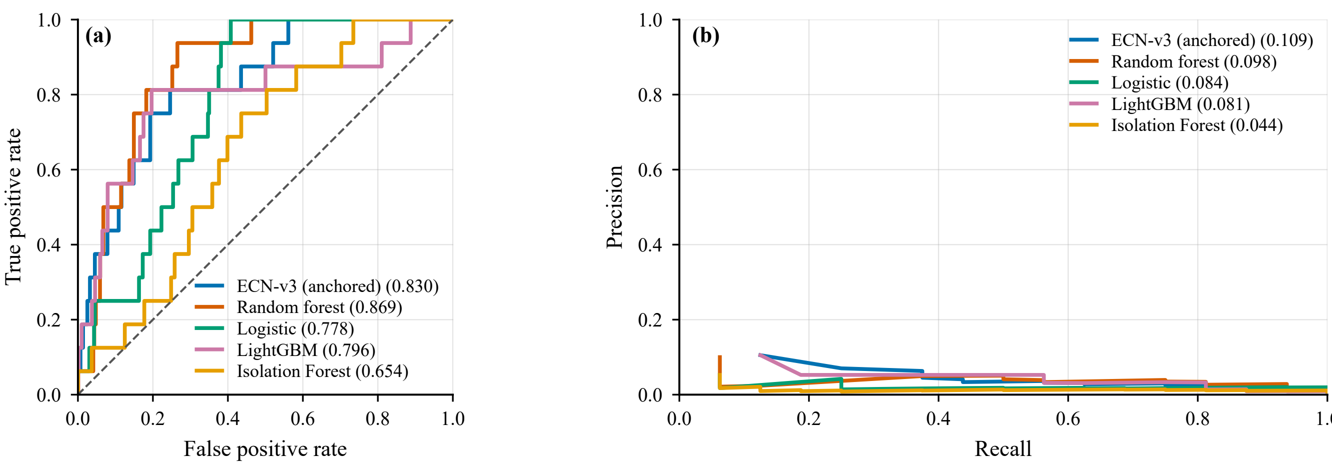

Fig. 4. T1 ROC (a) and precision–recall (b) curves on frozen instance v1.1.0-INST for representative detectors, including the final **anchored** ECN-v3 head. Legend values report ROC-AUC and average precision on this instance.

C. Calibration and operating points

Calibration quality is summarized in Table VIII. Final T1 raw Brier is 0.1509 (architecture selection), worse than RF (0.0227), motivating optional Beta calibration for probability displays without altering AUPRC ranking. Fig. 7 shows a reliability diagram on v1.1.0-INST for harness anchored ECN

TABLE V
PAIRED WILCOXON SIGNED-RANK TESTS AND CLIFF'S δ ON PER-SEED AUPRC. FOR T1, THE PROPOSED VECTOR IS THE ARCHITECTURE-SELECTION ANCHORED HEAD (`MANUSCRIPT_READY_NUMBERS.JSON`, MEAN 0.1152); BASELINES ARE FROM THE LATEST FULL-SUITE HARNESS. n=6 TESTS ARE UNDERPOWERED; NON-SIGNIFICANT p-VALUES MUST NOT BE OVER-CLAIMED.

Task	Baseline	p	Cliff δ	Proposed	Baseline
T1_anomaly	tabnet__full	0.062	0.667	0.1152	0.0492
	graphsage__full	0.031	0.944	0.1152	0.0152
	xgboost__full	0.094	0.722	0.1152	0.0471
	catboost__full	0.156	0.611	0.1152	0.0528
	lightgbm__full	0.062	0.611	0.1152	0.0570
	gradient_boosting__full	0.219	0.500	0.1152	0.0730
	balanced_rf__full	0.031	0.667	0.1152	0.0467
	random_forest__full	0.688	0.389	0.1152	0.0758
	logistic__full	0.031	0.556	0.1152	0.0631
	isolation_forest__full	0.031	0.667	0.1152	0.0575
	ewma__full	0.062	0.778	0.1152	0.0403
	threshold__full	0.031	0.778	0.1152	0.0315
	mlp_sequence__full	0.031	1.000	0.1152	0.0084
	gnn_graphsage_proxy__full	0.062	0.778	0.1152	0.0335
	majority__full	0.031	1.000	0.1152	0.0098
T2_failure	logistic__full	0.844	−0.500	0.0153	0.0380
	random_forest__full	0.688	−0.333	0.0153	0.0176

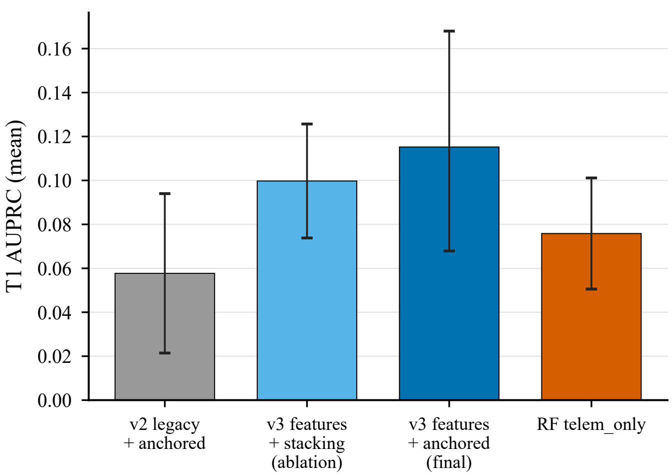

Fig. 5. T1 architecture selection: v2 legacy+anchored, v3+stacking (ablation), v3+anchored (final), and RF telem_only. Error bars denote 95% confidence intervals over six seeds.

TABLE VI
ABLATION STUDY OF MEAN AUPRC ON T1 AND T2 FROM THE LATEST FULL-SUITE HARNESS (SIX SEEDS, 95% CI). THE `FULL` ROW IS A HARNESS RE-RUN AND MAY DIFFER SLIGHTLY FROM THE ARCHITECTURE-SELECTION FINAL T1 MEAN (0.1152) REPORTED IN TABLE IV.

Task	Ablation	Mean	95% CI
T1_anomaly	full	0.1106	[0.0416, 0.1795]
	no_twin	0.1149	[0.0429, 0.1869]
	no_nbr	0.1159	[0.0513, 0.1805]
	telem_only	0.0686	[0.0145, 0.1227]
	twin_only	0.0283	[0.0129, 0.0436]
T2_failure	full	0.0153	[0.0043, 0.0262]
	no_twin	0.0140	[0.0028, 0.0252]
	no_nbr	0.0140	[0.0028, 0.0252]
	telem_only	0.0359	[−0.0066, 0.0783]
	twin_only	0.0156	[0.0082, 0.0230]

scores.

TABLE VII
ESTIMATED MODULE AUPRC CONTRIBUTION FROM THE FULL-SUITE HARNESS, DEFINED AS AUPRC(FULL)−AUPRC(ABLATED). POSITIVE VALUES INDICATE IMPROVEMENT WHEN THE MODULE IS PRESENT. THESE HARNESS DELTAS ARE DISTINCT FROM THE ARCHITECTURE-SELECTION TWIN GAIN UNDER THE FINAL HEAD ($\approx +0.00035$) SHOWN IN FIG. 6.

Task	Telemetry	Neighbor	Digital Twin
T1_anomaly	0.0823	−0.0054	−0.0043
T2_failure	−0.0004	0.0013	0.0013

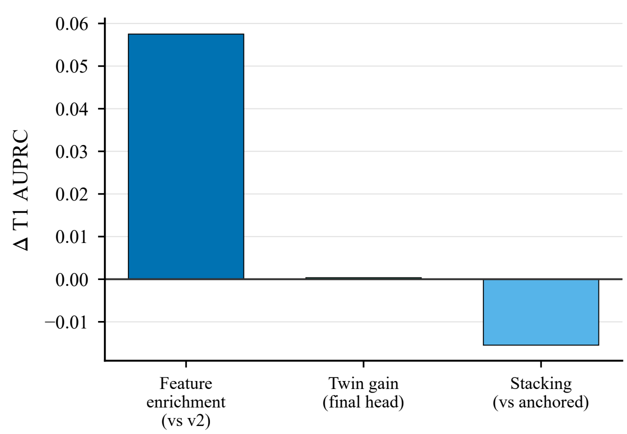

Fig. 6. Verified T1 AUPRC deltas from architecture selection: feature enrichment versus v2, twin gain under the final head, and stacking versus anchored fusion.

D. RCA, healing decision support, and cost

Table IX reports RCA macro-F1, healing decision-support scores, and cost. Proposed T3 RCA macro-F1 mean is 0.2540 using RF with TreeSHAP explanations. Healing with RCA category features reaches macro-F1 1.0 on this synthetic setup—an upper bound that is *not* interpreted as operational perfection; the honest `no_rca_cat` setting yields 0.1903. Fig. 8 illustrates a representative T3 confusion matrix on the primary instance.

TABLE VIII
CALIBRATION QUALITY MEASURED BY BRIER SCORE (MEAN OVER SIX SEEDS WITH 95% CI). LOWER IS BETTER. THE T1 ECN-V3 FINAL ROW USES ARCHITECTURE-SELECTION BRIER (MANUSCRIPT_READY_NUMBERS.JSON); REMAINING ROWS USE THE FULL-SUITE HARNESS.

Task	Method	Mean	95% CI
T1_anomaly	ECN-v3 final (selection)	0.1509	[0.0928, 0.2090]
T1_anomaly	Random forest	0.0227	[0.0112, 0.0342]
T1_anomaly	Logistic	0.2328	[0.1742, 0.2915]
T2_failure	ECN proposed	0.2089	[0.1190, 0.2987]
T2_failure	Logistic	0.2577	[0.2079, 0.3074]
T2_failure	Random forest	0.0124	[0.0087, 0.0160]

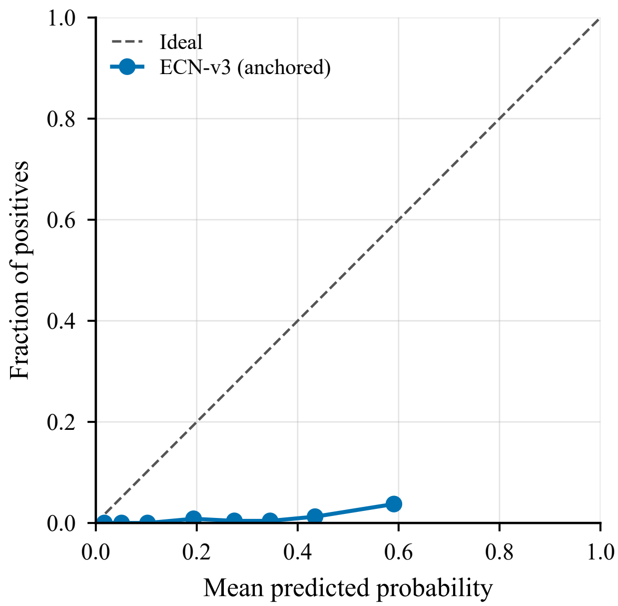

Fig. 7. T1 reliability diagram on instance v1.1.0-INST for the anchored ECN-v3 head. The dashed line denotes perfect calibration. Optional Beta calibration is recommended for operator-facing probabilities.

TABLE IX
RCA ACCURACY, HEALING DECISION-SUPPORT SCORES, AND EVALUATION WALL-CLOCK COST (SIX SEEDS). THE FULL HEALING SETTING INCLUDES RCA CATEGORY FEATURES; NO_RCA_CAT IS THE HONEST TELEMETRY-ORIENTED BASELINE.

Metric	Setting	Mean [95% CI]
T3 RCA macro-F1	proposed	0.2540 [0.0901, 0.4179]
Healing macro-F1	full (w/ RCA cat.)	1.0000 [1.0000, 1.0000]
Healing macro-F1	no_rca_cat	0.1903 [0.0989, 0.2816]
Eval. wall time (s/seed)	full suite	119.69 [111.76, 127.61]

E. Why simplicity remains preferable

At ∼1% positives, enriched telemetry recovers most ranking signal. Anchored fusion is simpler than stacking, improves mean T1 AUPRC on the same features, and remains competitive with RF in mean while acknowledging non-significant paired tests versus RF. With n=6 seeds, intervals are wide; claims emphasize reproducible mean improvements and architecture honesty rather than universal paired dominance. Deep tabular (TabNet) and true GraphSAGE message passing did not displace RF or ECN-v3 on mean T1 AUPRC under this protocol (Section VI-F).

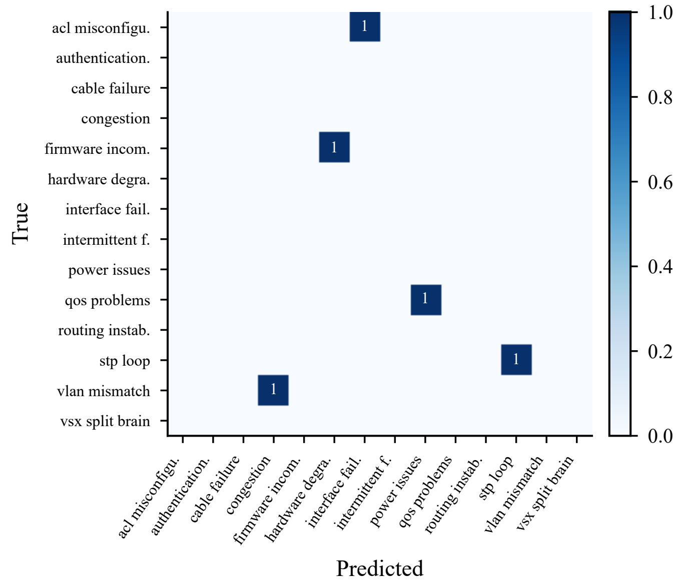

Fig. 8. Representative T3 RCA confusion matrix for the RF+TreeSHAP RCA head on instance v1.1.0-INST.

TABLE X
DEEP BASELINES ON FROZEN SIX-SEED T1/T2. ECN-V3 FINAL USES ARCHITECTURE-SELECTION NUMBERS (MANUSCRIPT_READY_NUMBERS.JSON, BOOTSTRAP 95% CI). TABNET AND TRUE GRAPHSAGE USE THE FULL-SUITE HARNESS. THE HISTORICAL GNN PROXY IS LIGHTGBM, NOT MESSAGE-PASSING.

Method	T1 AUPRC	T2 AUPRC
ECN-v3 final (manuscript)	0.1152 [0.0679, 0.1680]	—
TabNet	0.0492 [0.0213, 0.0772]	0.0110 [0.0038, 0.0183]
GraphSAGE (true GNN)	0.0152 [0.0059, 0.0246]	0.0062 [0.0057, 0.0068]
GNN proxy (LightGBM)	0.0335 [0.0167, 0.0504]	0.0106 [0.0069, 0.0144]
Random forest (telem)	0.0758 [0.0427, 0.1090]	0.0176 [0.0099, 0.0253]

F. Deep tabular and true GNN baselines

Table X reports measured six-seed means for TabNet and true GraphSAGE, alongside the historical GNN proxy and RF. Artifacts: `results/aggregate_v3.json` keys `tabnet__full`, `graphsage__full`, `gnn_graphsage_proxy__full`; manuscript pointer `extensions_v5_deep_baselines` in `results/manuscript_ready_numbers.json`.

On T1, TabNet attains mean AUPRC 0.0492 (95% CI $[0.0213, 0.0772]$), below RF (0.0758) and far below the manuscript-final ECN-v3 head (0.1152). True GraphSAGE attains 0.0152 (95% CI $[0.0059, 0.0246]$) with 19 devices and 54 directed edges in the twin graph—competitive with neither RF nor ECN on this rare-event ranking task. The historical GNN proxy remains at 0.0335 and must not be interpreted as message-passing performance. On T2, TabNet (0.0110) and GraphSAGE (0.0062) also trail telem logistic (0.0380). These outcomes support reporting deep baselines for completeness while retaining the evidence-selected ECN-v3 / telem-logistic heads.

VII. Discussion

A. What to claim

The defensible claims are architectural and methodological: (1) ECNetBench provides a leakage-aware multi-task enterprise benchmark with multi-seed frozen instances; (2) ECN-v3 closes the loop from detection to healing *decision support* with TreeSHAP RCA; (3) leakage-safe temporal enrichment is the primary driver of the T1 AUPRC lift from 0.0577 (v2) to 0.1152 (final); (4) anchored fusion is preferred over stacking on this benchmark (higher mean AUPRC, lower complexity); (5) T2 remains best served by a telem logistic head under extreme rarity.

B. What not to claim

We do not claim statistically significant superiority over random forest on T1 (p=0.688), significant superiority of anchored over stacking (p=0.375), live deployment on physical AOS-CX switches, or that multi-agent fusion alone beats strong tabular detectors without feature enrichment. We also do not claim that TabNet or true GraphSAGE outperform RF/ECN on ECNetBench: measured mean T1 AUPRC is 0.0492 (TabNet) and 0.0152 (GraphSAGE) versus 0.0758 (RF) and 0.1152 (final ECN-v3). Impact AUPRC of 1.0 for some models on this synthetic label construction is reported for completeness but interpreted cautiously as potential label/feature collinearity rather than solved service-impact prediction in the wild.

C. Practical NetOps implications

Operators can use ECN as a sandbox for ranking alerts, proposing RCA categories with SHAP attributions, and suggesting remediation classes before human approval. Prefer the anchored head for T1 triage; prefer telem logistic scores for T2 horizon ranking; apply optional Beta calibration when displaying probabilities.

D. Governance and operational implications

Because ECN-v3 closes a multi-agent loop that can recommend remediation classes, deployment in enterprise environments should retain human approval gates for delegated actions, consistent with governance analyses of agentic systems and agent-to-agent delegation [21], [22], [28]. Leakage-safe temporal evaluation and checksummed instances support operational-data auditability and trust assessment [24], while cybersecurity risk-control and zero-exposure guidance caution against exposing privileged recovery or identity pathways through automated tooling [23], [25]. If generative or retrieval-augmented assistants are later attached to NetOps workflows, enterprise generative-AI security and agentic-AI governance frameworks provide complementary controls beyond detector metrics [26], [27]. Finally, semantic tracing and observability practices developed for LLM-based multi-agent stacks offer a useful template for explaining and governing agent decisions alongside TreeSHAP RCA [29]. Digital-transformation studies further remind operators that legacy constraints and human resistance often dominate outcomes even when models improve ranking scores [4]–[7].

TABLE XI
Threats to validity and corresponding mitigations.

Threat	Mitigation / residual risk
Synthetic data	Realism audits; still not a substitute for production traces.
External validity	AOS-CX-style assumptions; other vendors/fabrics may differ.
Label leakage	Temporal freeze and feature audits; healing full vs. `no_rca_cat`.
Small n seeds	Six seeds with confidence intervals; tests remain underpowered.
Topology scale	19 devices / 31 links; scalability to large fabrics unproven.
Healing semantics	Decision support only; no live actuation evidence.

VIII. Limitations and Threats to Validity

Table XI summarizes the principal threats to validity. Construct validity risks include synthetic incident generators that may over-structure causal chains. Internal validity is strengthened by frozen instances, checksums, and relative-path reproduction. Conclusion validity is limited by n=6 and wide confidence intervals. External validity requires future evaluation on anonymized production telemetry.

IX. Conclusion

We presented ECNetBench and an Enterprise Cognitive Network (ECN-v3) digital-twin multi-agent framework evaluated under leakage-safe multi-seed protocols. Verified means show final T1 AUPRC 0.1152 for leakage-safe enriched features with anchored fusion, exceeding random forest 0.0758 and the v2 configuration 0.0577. Stacking on the same enriched features is a negative ablation (0.0997). Added deep baselines—TabNet (0.0492) and true GraphSAGE (0.0152)—do not overturn these selections. The recommended T2 head is telemetry logistic regression (0.0380). RCA uses RF with TreeSHAP. The primary scientific value is a reproducible benchmark plus a complete cognitive NetOps pipeline with evidence-selected fusion and honest statistics—not a universal detector champion. Future work includes larger topologies, production-trace calibration, richer temporal GNNs, human-in-the-loop healing studies, and stronger governance-oriented observability (e.g., semantic tracing of multi-agent decision paths) under enterprise AI security controls [26]–[29].

Appendix A
Supplementary Notes

A. Reproduction commands

```
python -m venv .venv
pip install -r requirements.txt
python scripts/download_instances.py
python scripts/verify_instances.py
pytest -q
python evaluation/trace_t1_gain.py
python evaluation/select_t1_architecture.py
python evaluation/run_full_evaluation.py
python paper/overleaf/scripts/\
regenerate_pub_figures_v3.py
pdflatex main.tex
bibtex main
pdflatex main.tex
pdflatex main.tex
```

B. Traceability

Final T1 numbers map to `results/manuscript_ready_numbers.json` and `results/v3_gated/t1_architecture_selection.json`. Baseline/T2/RCA aggregates map to `results/aggregate_v3.json` and `results/per_seed/*.json`. Additional reports: `reports/FINAL_ARCHITECTURE_SELECTION.md`, `reports/T1_IMPROVEMENT_TRACEABILITY.md`. The Overleaf project is synchronized under `paper/overleaf/`.

C. Additional figures

Per-seed ROC/PR, confusion matrices, and calibration plots for T1/T2/T3 are included under `figures/` as vector PDF and 600 dpi PNG pairs. T2 curves and the T1 confusion matrix for the primary instance are shown below; remaining seeds are archived for completeness.

Data Availability

Frozen ECNetBench SQLite instances and checksums will be released with the camera-ready version of this work. For double-blind review, instances are provided via an anonymized supplementary archive whose contents match relative path `benchmark/instances/` and checksums in `benchmark/INSTANCE_CHECKSUMS.json` (SHA-256 digests; reviewers should verify the archive against that manifest). Evaluation scripts read instances only through relative paths.

Code Availability

Source code for the generator, Digital Twin, agents, baselines, ablations, and evaluation harness will be released under an MIT license with the camera-ready version. An anonymized code archive matching the submitted artifact hash is provided for review; public repository URLs are withheld during double-blind review and restored upon acceptance. The Overleaf manuscript source is synchronized under `paper/overleaf/`. Exact reproduction commands appear in the companion reproducibility notes.

Author Contributions

Conceptualization, methodology, software, validation, investigation, data curation, writing—original draft, and visualization were performed by the author(s). Formal analysis and statistical testing follow the published evaluation scripts.

Conflict of Interest

The authors declare no competing interests.

Acknowledgments

[Acknowledgments to be completed upon acceptance.]

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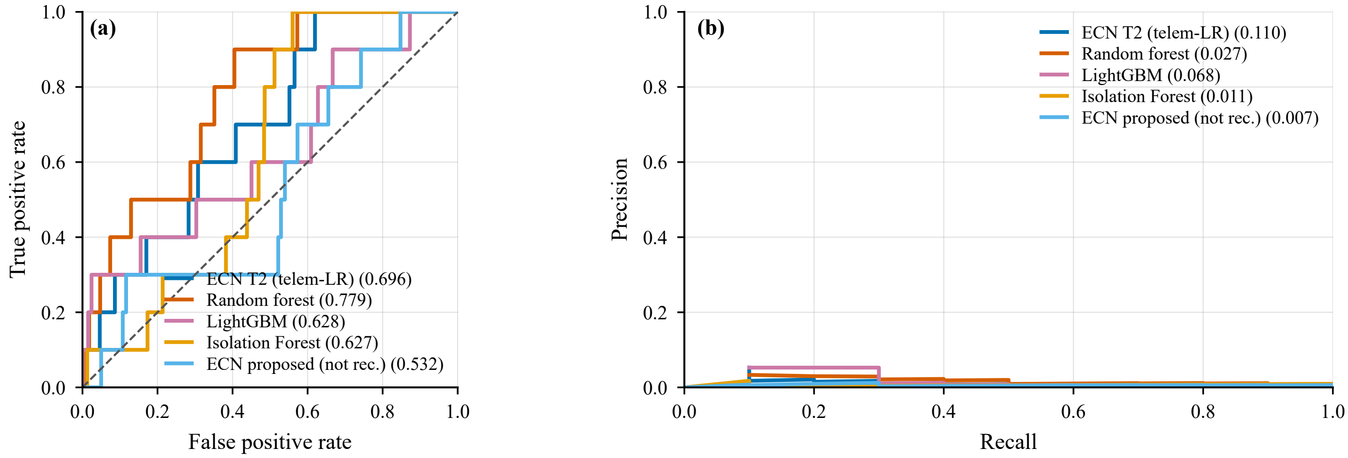

Fig. 9. T2 ROC (a) and precision–recall (b) curves on frozen instance v1.1.0-INST. The recommended T2 operating head is telemetry logistic regression (labeled ECN T2 (telem-LR)); the multi-agent proposed T2 curve is shown only for contrast.

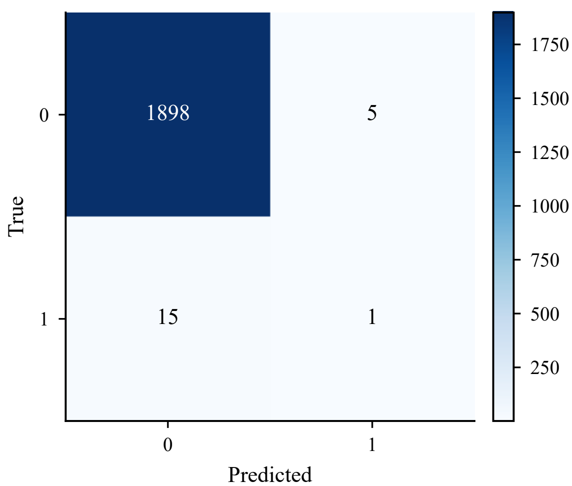

Fig. 10. T1 confusion matrix on instance v1.1.0-INST at a validation-tuned threshold (threshold unused for AUPRC).

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